Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: Chan Chi Yuen

Student ID: 24128204D

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0 via R1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? No 0

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you? 2

It shows that when all the interfaces on R2 are passive, R2 stops sending OSPF Hello packets and no longer forms adjacencies. Thus, its networks are not advertised to the neighbors

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)

R2(config-router) 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost is 65 2

Is R2 listed as an OSPF neighbor to R3? Yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

__To ensure the cost calculations on high-speed interfaces are accurate. After the change, the OSPF can_ differentiate properly between FastEthernet, GigabitEthernet, and higher-speed links.__________________ 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24 = 64 ____

Cost to 192.168.23.0/30 = 128 ____ 0

____

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

__The new accumulated cost is 1562. It is because the OSPF cost will increase when lowering the___ 2 interface bandwidth (reference bandwidth ÷ interface bandwidth).________________________________

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

__It is because the link cost from R1 to R3 increases after changing the bandwidth. OSPF is recalculated and found that the path that goes through R2 is a lower cost. Thus, the traffic is routed via R2._________ 4

____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   To ensure that the identification of routers in the OSPF domain is predictable and consistent. It also simplifies the troubleshooting and avoids conflicts.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   It is because the point-to-point serial links are used in this lab. The DR/BDR elections are not needed.

   4

3. Why would you want to set an OSPF interface to passive?

   It is because it can prevent the unnecessary OSPF Hello packets on LAN interfaces where no other routers exist. It can reduce overhead.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of area allows hierarchical design and scalability. For example, the backbone area 0 and other areas.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The OSPF route will be used because OSPF has the lowest administrative distance (110), compared to RIP (120) and EIGRP (90)

   EIGRP 0